

LECTURE NOTES FOR SURV 616

Statistical Methods II

Categorical Data Analysis

CLASS #10

OVERVIEW: In this class we will continue our discussion of the statistical properties of categorical data from a multinomial distribution. We will discuss properties of 2-way tables, along with measures of association. In addition, we will discuss tests of independence.

1. Further Results for Multinomial Distributions

In this section we present some additional material on multinomial distributions that was not covered in the previous class. We begin by discussing the distribution of combinations of individual cells in a multinomial table

Distribution of Collapsed Multinomials

Let $\mathbf{Y}' = (Y_1, \dots, Y_k)$ be a k dimensional multinomial random vector with parameters $(n, \mathbf{p})$. We saw from the previous class that

$$(1.1) \quad Y_i \sim \text{Bin}(n, p_i) \quad \text{for} \quad i = 1, \dots, k$$

An important question is “What is the distribution of $Y_1 + Y_2$, and the new $k - 1$ vector $\mathbf{Y}_*$ ” where

$$(1.2) \quad \mathbf{Y}'_* = (Y_1 + Y_2, Y_3, \dots, Y_k)$$

It turns out that

$$(1.3) \quad Y_1 + Y_2 \sim \text{Bin}(n, p_1 + p_2)$$

$$\mathbf{Y}'_* \sim \text{Mult}_{k-1}(n, \mathbf{p}_*)$$

$$\mathbf{p}'_* = (p_1 + p_2, p_3, \dots, p_k)$$

The result seems intuitively obvious, but in fact the result is not as trivial as one might think. For example, it is not generally the case that

$$(1.4) \quad \text{NOT USUALLY THE CASE} \quad \text{Bin}(n, p_1) + \text{Bin}(n, p_2) \sim \text{Bin}(n, p_1 + p_2)$$

but it is when we combine cells of a multinomial random vector. We will not give a formal proof of the result in (1.3), but a natural way of proving it is to think of the multinomial as a sum of n trials and collapse the indicator vectors for the individual trials.

This distributional result will be useful later when we need the distribution of marginal sums in 2-Way and Higher-Way tables.

Goodness-of-fit Tests with estimated Parameters

In section 3 of the last class we looked at testing the set of multinomial probabilities against a specified hypothesis, and constructed the Pearson X^2 and the Likelihood ratio G^2 statistic. In both cases, under the null hypothesis, the statistics had an asymptotic Chi-square distribution with $k - 1$ degrees-of-freedom when we are dealing with a k dimensional multinomial random vector.

In practice we are often interested in testing hypothesis where the null hypothesis specifies that the multinomial probability vector is a function of a small set ($< k - 1$) of parameters which are themselves unspecified. This is commonly the case when we test whether a sample comes from a particular parametric distribution, and we have a sample of size n . In this case we can still use the Pearson X^2 and the Likelihood ratio G^2 statistic, where they are computed in their *Observed and Expected* form, but now they have an asymptotic Chi-square distribution with degrees-of-freedom given by

$$(1.5) \quad df = k - 1 - (\text{number of free parameters})$$

The number of free parameters is determined by H_0 . We illustrate this with the next example.

Example 1. (Taken from **Introduction to Probability Theory and Statistical Inference** by Harold Larson). Assume that $X_1, \dots, X_{300}$ is an independent random sample of yardsticks from a yardstick manufacturer and it is assumed that the lengths are all independently distributed with a $N(\mu, \sigma^2)$ distribution. Assume the sample satisfies

Yardstick Lengths (in inches)	
True Length	Frequency
$35.7 < X_i \leq 35.8$	10
$35.8 < X_i \leq 35.9$	30
$35.9 < X_i \leq 36.0$	104
$36.0 < X_i \leq 36.1$	97
$36.1 < X_i \leq 36.2$	51
$36.2 < X_i \leq 36.3$	8
	300

Assume that

$$\hat{\mu} = 36.008 \quad \hat{\sigma}^2 = 0.0114$$

then we can estimate the probability of falling into each of the intervals above. For example

$$P\{36.2 < X_i \leq 36.3\} = \Phi\left(\frac{36.3 - \hat{\mu}}{\hat{\sigma}}\right) - \Phi\left(\frac{36.2 - \hat{\mu}}{\hat{\sigma}}\right)$$

We obtain

$$\begin{array}{lll} p_1 = 0.0262 & E_1 = 7.86 & O_1 = 10 \\ p_2 = 0.1300 & E_2 = 39.0 & O_2 = 30 \\ p_3 = 0.3159 & E_3 = 94.77 & O_3 = 104 \\ p_4 = 0.3330 & E_4 = 99.99 & O_4 = 97 \\ p_5 = 0.1582 & E_5 = 47.46 & O_5 = 51 \\ p_6 = 0.0367 & E_6 = 11.01 & O_6 = 8 \end{array}$$

$$\chi^2 = 4.73 \quad \chi^2_3(0.05) = 7.81$$

$$G^2 = 4.74 = 2 \sum_{i=1}^k O_i \ln \left(\frac{O_i}{E_i} \right)$$

$$k = 5 \quad df = 6 - 1 - 2 = 3$$

Under $H_0 : P_1 = P_6, P_2 = P_5, P_3 = P_4$ there are 3 parameter values but since they must sum to 1, we have 2 free parameters. Both G^2 and X^2 are distributed asymptotically as Chi-square random variables with 3 degrees-of-freedom. Both statistics lead us to accept the null hypothesis that the observed yardstick lengths are normally distributed.

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Sometimes we are interested in testing hypotheses about the multinomial probability in a less parametric way, such as testing that some of the multinomial probabilities are the same. We illustrate this example in the next example which tests *symmetry* of the multinomial probabilities.

Example 2. Consider a random sample of $n = 3343$ Swedish families with 4 or more children, taken from Edwards and Fraccaro (1960). The number of *boys* among the first 4 children was recorded below

Data from Swedish Families					
Category Number	1	2	3	4	5
Number of Boys	0	1	2	3	4
Observed Frequency	183	789	1250	875	246

Let p_i be the probability for the i -th category. We want to test the hypothesis that

$$H_0 : p_1 = p_5, p_2 = p_4 \text{ versus } H_A : p_1 \neq p_5 \text{ or } p_2 \neq p_4$$

which is the test of symmetry in boys and girls since $p_1 = p_5$ says that the probability of having 4 girls is the same as having 4 boys, and $p_2 = p_4$ says that the probability of having 3 girls is the same as having 3 boys. Under the null hypothesis we can write

$$p_1 = p_5$$

$$p_2 = p_4$$

$$p_3 = 1 - p_1 - p_2 - p_4 - p_5 = 1 - 2p_1 - 2p_2$$

$$p_4 = p_2$$

$$p_5 = p_1$$

which says that we have two free parameters to estimate. So we can write the likelihood function under the null hypothesis as

$$L(p_1, p_2; \mathbf{n}) = \frac{n!}{n_1! n_2! n_3! n_4! n_5!} p_1^{n_1} p_2^{n_2} (1 - 2p_1 - 2p_2)^{n_3} p_2^{n_4} p_1^{n_5}$$

Then we can write

$$\begin{aligned} \ln[L(p_1, p_2; \mathbf{n})] &= \ln\left(\frac{n!}{n_1! n_2! n_3! n_4! n_5!}\right) + (n_1 + n_5) \ln(p_1) \\ &\quad + (n_2 + n_4) \ln(p_2) + n_3 \ln(1 - 2p_1 - 2p_2) \end{aligned}$$

and

$$\begin{aligned} \frac{\partial}{\partial p_1} \ln[L(p_1, p_2; \mathbf{n})] &= \frac{n_1 + n_5}{p_1} - \frac{2n_3}{1 - 2p_1 - 2p_2} \\ \frac{\partial}{\partial p_2} \ln[L(p_1, p_2; \mathbf{n})] &= \frac{n_2 + n_4}{p_2} - \frac{2n_3}{1 - 2p_1 - 2p_2} \end{aligned}$$

So we obtain

$$\hat{p}_1 = (2n)^{-1} (n_1 + n_5)$$

$$\hat{p}_2 = (2n)^{-1} (n_2 + n_4)$$

and

$$\hat{p}_4 = \hat{p}_2 = (2n)^{-1} (n_2 + n_4)$$

$$\hat{p}_5 = \hat{p}_1 = (2n)^{-1} (n_1 + n_5)$$

$$\hat{p}_3 = 1 - 2\hat{p}_1 - 2\hat{p}_2 = 1 - n^{-1} (n_1 + n_5) - n^{-1} (n_2 + n_4) = n^{-1} n_3$$

Therefore

$$\begin{array}{ll}
E_1 = n\hat{p}_1 = \frac{1}{2}(n_1 + n_5) = 214.5 & O_1 = n_1 = 183 \\
E_2 = n\hat{p}_2 = \frac{1}{2}(n_2 + n_4) = 832 & O_2 = n_2 = 789 \\
E_3 = n\hat{p}_3 = n_3 = 1250 & O_3 = n_3 = 1250 \\
E_4 = n\hat{p}_4 = \frac{1}{2}(n_2 + n_4) = 832 & O_4 = n_4 = 875 \\
E_5 = n\hat{p}_5 = \frac{1}{2}(n_1 + n_5) = 214.5 & O_5 = n_5 = 246
\end{array}$$

We obtain

$$X^2 = \sum_{i=1}^5 \frac{(O_i - E_i)^2}{E_i} = 13.70$$

$$G^2 = 2 \sum_{i=1}^5 O_i \ln \left(\frac{O_i}{E_i} \right) = 13.73$$

Both statistics have an asymptotic Chi-square distribution with degrees-of-freedom given by

$$df = (5-1) - (p_1 + p_2) = (5-1) - 2 = 2$$

Since $\chi_2^2(0.01) = 9.21$ we reject the null hypothesis.

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2. Analysis of 2-Way Tables

In this section we discuss 2-Way tables, also called 2-dimensional tables. In some cases we limit our discussion to 2×2 tables and in other cases we discuss the more general $I \times J$ case. We begin by discussing general measures of association in 2×2 tables, then discuss testing for independence in the general $I \times J$ table case, and finally discuss the analysis of ordinal data in the $I \times J$ table case. In all cases we will assume that the tables correspond to multinomial sampling.

2.1 Measures of Association

We will limit our discussion of measures of association to 2×2 tables in order to simplify the discussion. In our discussion we will assume the following 2×2 table of counts

Row	Column	
	1	2
1	n_{11}	n_{12}
2	n_{21}	n_{22}

and the following table of probabilities

Row	Column	
	1	2
1	π_{11}	π_{12}
2	π_{21}	π_{22}

We will consider the column as the response variable, and the row as the independent (or predictor) variable. Define the conditional probabilities

$$(2.1.1) \quad \pi_{j|i} = \frac{\pi_{ij}}{\pi_{i+}}$$

where

$$\pi_{i+} = \pi_{i1} + \pi_{i2}$$

The conditional probability $\pi_{j|i}$ is the probability of falling in the j -th column given that you are in the i -th row.

Relative Risk

We then define the *relative risk* as the ratio

$$(2.1.2) \quad \text{Relative Risk (Response Category 1)} = \frac{\pi_{1|1}}{\pi_{1|2}} = \frac{\Pr(D|M)}{\Pr(D|F)}$$

Similarly one can define a relative risk for response category 2 as

$$(2.1.3) \quad \text{Relative Risk (Response Category 2)} = \frac{\pi_{2|1}}{\pi_{2|2}} = \frac{\Pr(\bar{D}|M)}{\Pr(\bar{D}|F)}$$

The relative risk is heavily used in Epidemiology, and we will discuss it more later. It is a nonnegative real number. We can estimate the relative risk in (2.1.2) by

$$(2.1.4) \quad \frac{n_{11} / n_{1+}}{n_{21} / n_{2+}}$$

$$\hat{V} \left\{ \ln \left(\frac{n_{11} / n_{1+}}{n_{21} / n_{2+}} \right) \right\} = \frac{1 - \frac{n_{11}}{n_{1+}}}{n_{11}} + \frac{1 - \frac{n_{21}}{n_{2+}}}{n_{21}} = \frac{\Pr(\bar{D}|M)}{n_{11}} + \frac{\Pr(\bar{D}|F)}{n_{21}}$$

Where we have also given an estimator of the *log-relative risk*. We can do a similar thing for the relative risk in (2.1.3) by

$$(2.1.5) \quad \frac{n_{12} / n_{1+}}{n_{22} / n_{2+}}$$

$$\hat{V} \left\{ \ln \left(\frac{n_{12} / n_{1+}}{n_{22} / n_{2+}} \right) \right\} = \frac{1 - \frac{n_{12}}{n_{1+}}}{n_{12}} + \frac{1 - \frac{n_{22}}{n_{2+}}}{n_{22}} = \frac{\Pr(D|M)}{n_{11}} + \frac{\Pr(D|F)}{n_{21}}$$

Odds (or Cross-Product) Ratio

Within row 1 the *odds* that the response is in column 1 instead of column 2 is defined as

$$(2.1.6) \quad \text{Odds}_1 = \frac{\pi_{1|2}}{\pi_{2|1}} = \frac{\pi_{11}}{\pi_{12}} \quad \frac{\Pr(D|M)}{\Pr(\bar{D}|M)} = \frac{\Pr(D|M)}{1 - \Pr(D|M)}$$

while the odds for row 2 is

$$(2.1.7) \quad Odds_2 = \frac{\pi_{1|2}}{\pi_{2|2}} = \frac{\pi_{21}}{\pi_{22}} \quad \frac{\Pr(D|F)}{\Pr(\bar{D}|F)} = \frac{\Pr(D|F)}{1 - \Pr(D|F)}$$

The ratio of the odds

$$(2.1.8) \quad \theta = \frac{\pi_{1|1} / \pi_{2|1}}{\pi_{1|2} / \pi_{2|2}} = \frac{\pi_{11}\pi_{22}}{\pi_{21}\pi_{12}}$$

is called the *odds ratio*, or *cross-product ratio*. It is called the cross-product ratio because it is the ratio of the products of the cross diagonal elements in the 2×2 table.

The following description is taken from Agresti:

“The odds ratio can equal any non negative number. When all cell probabilities are positive, independence of the (*independent variable*) and the (*response variable*) is equivalent to $\theta = 1$. When $1 < \theta < \infty$, subjects in row 1 are more likely to make the first response than are subjects in row 2; that is $\pi_{1|1} > \pi_{1|2}$. For instance, when $\theta = 4$, the odds of the first response are four times higher in row 1 than in row 2. (This does not mean that the *probability* $\pi_{1|1}$ is four times higher than $\pi_{1|2}$; that is the interpretation of a *relative risk* of 4.0.) When $0 < \theta < 1$, the first response is less likely in row 1 than in row 2; that is $\pi_{1|1} < \pi_{1|2}$. When one cell has zero probability, θ equals 0 or ∞ .

The odds ratio does not change value when the orientation of the table is reversed so that the rows become the columns and the columns become the rows. Therefore, it is unnecessary to identify one classification as the response variable in order to calculate θ .

Values of θ farther from 1.0 in a given direction represent stronger levels of association. Two values for θ represent the same level of association, but in opposite directions, when one value is the inverse of the other. For instance, when $\theta = 0.25$, the odds of the first response are 0.25 times as high in row 1 as in row 2, or equivalently $(0.25)^{-1} = 4$ times as high in row 2 as in row 1. When the order of the rows is reversed or the order of the columns is reversed, the new value of θ is the inverse of the original value.

It is sometimes more convenient to use $\ln(\theta)$, the natural logarithm of θ . Independence corresponds to $\ln(\theta) = 0$. The log odds ratio is symmetric about this value – reversal of rows or of columns results in a change in its sign. Two values for $\ln(\theta)$ that are the same except for sign, such as $\ln(4) = 1.39$ and $\ln(0.25) = -1.39$, represent the same level of association.”

We can estimate the cross-product ratio by

$$(2.1.9) \quad \hat{\theta} = \frac{n_{11}n_{22}}{n_{21}n_{12}}$$

The variance of $\ln(\hat{\theta})$ can be estimated by

$$(2.1.10) \quad \hat{V}\{\ln(\hat{\theta})\} = \frac{1}{n_{11}} + \frac{1}{n_{12}} + \frac{1}{n_{21}} + \frac{1}{n_{22}}$$

We illustrate these concepts in the next example.

Example 3. The following example is taken from the New England Journal of Medicine (1988) **318** pages 262-262. The data is from a Physicians’ Health Study in which every day physicians participating in the study took either one aspirin tablet or a placebo. The data are presented below

Physicians’ Health Study Data				
	Myocardial Infraction		Total	P_{it}
	Heart Attack	No Attack		
Placebo	189	10845	11034	0.017
Aspirin	104	10933	11037	0.009

Here we treat Myocardial Infraction as the response variable and Placebo/ Aspirin as the independent variable. The total number of physicians given the placebo was

$$n_{1+} = 11034$$

and those given aspirin was

$$n_{2+} = 11037$$

The relative risk of a heart attack is given by

$$\text{Relative risk of Heart Attack} = \frac{189/11034}{104/11037} = 1.82$$

Therefore the proportion suffering heart attacks was 1.82 times higher for those taking the placebo than for those taking aspirin. The estimated cross-product ratio and log-cross product ratio is given by

$$\hat{\theta} = \frac{(189)(10933)}{(104)(10845)} = 1.83$$

$$\ln(\hat{\theta}) = 0.60543771$$

with

$$\hat{V}\{\ln(\hat{\theta})\} = \frac{1}{189} + \frac{1}{104} + \frac{1}{10845} + \frac{1}{10933} = 0.01509006$$

$$\sqrt{\hat{V}\{\ln(\hat{\theta})\}} = 0.12284$$

Notice that the variance is determined by the smallest n_{ij} , implying that rare characteristics have higher variances. So we would report an approximate 95% confidence interval for $\ln(\theta)$ as

$$(0.364, 0.846) \quad (\text{CI on Odds Ratio is } (1.440, 2.331) .)$$

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Relationship between Relative Risk and the Odds Ratio

In the last example we saw the estimated relative risk and the estimated cross-product were very close. This was not an accident. Using (2.1.2) and (2.1.8) we can write

$$(2.1.11) \quad \theta = \text{Relative Risk (Response Category } i) \left(\frac{1 - \pi_{i|2}}{1 - \pi_{i|1}} \right)$$

Therefore

$$(2.1.12) \quad \theta = \text{Relative Risk (Response Category } i) \quad \text{when} \quad \pi_{i|1} \approx \pi_{i|2} \approx 0$$

This is precisely the case in Example 3 since heart attacks are a rare occurrence. We will return to this subject again when we cover epidemiological methods.

Measures of Association for $I \times J$ Tables

The measures of association were all described for the 2×2 table case. In the general $I \times J$ and higher dimension tables it is possible to talk about cross product ratio, but we will prefer to parameterize those higher dimensional tables with log-linear models which we will discuss later.

2.2 Testing Independence/ Homogeneity

Perhaps the most widely constructed test for 2-Way tables is the test for independence. Our general set up will be that we have a table of counts obtained by multinomial sampling given by

$$(2.2.1) \quad n_{ij} \quad \text{for} \quad i = 1, \dots, I, j = 1, \dots, J$$

with cell probabilities given by

$$(2.2.2) \quad \pi_{ij} \quad \text{for} \quad i = 1, \dots, I, j = 1, \dots, J$$

where i denotes row and j denotes column. We denote the marginal probabilities for rows by

$$(2.2.3) \quad \pi_{i+} \quad \text{for} \quad i = 1, \dots, I$$

and the marginal probabilities for columns by

$$(2.2.4) \quad \pi_{+j} \quad \text{for} \quad j = 1, \dots, J$$

Then the hypothesis of Independence which is being tested is

$$(2.2.5) \quad H_0 : \pi_{ij} = \pi_{i+}\pi_{+j} \text{ for all } i, j \text{ versus } H_A : \pi_{ij} \neq \pi_{i+}\pi_{+j} \text{ for some } i, j$$

Note that

$$(2.2.6) \quad \pi_{ij} = \pi_{i+}\pi_{+j} \Rightarrow \Pr ob\{Row = i | Column = j\} = \frac{\pi_{ij}}{\pi_{+j}} = \pi_{i+}$$

which says that the conditional distribution of row category probabilities is the same within each column. Therefore the conditional distributions are said to be *homogeneous* since they are the same across columns. Also note that

$$(2.2.7) \quad \pi_{ij} = \pi_{i+}\pi_{+j} \Rightarrow \Pr ob\{Column = j | Row = i\} = \frac{\pi_{ij}}{\pi_{i+}} = \pi_{+j}$$

which says that conditional distribution of column category probabilities is the same within each row. Likewise the conditional distributions are homogeneous since they are the same across rows. Therefore, because of both (2.2.6) and (2.2.7) the test of independence is also called a test of homogeneity.

We can write the likelihood function under the null hypothesis as

$$(2.2.8) \quad L(\pi; \mathbf{n}) = \frac{n!}{\prod_{i=1}^I \prod_{j=1}^J n_{ij}!} \prod_{i=1}^I \prod_{j=1}^J (\pi_{i+} \pi_{+j})^{n_{ij}}$$

and the log-likelihood function as

$$(2.2.9) \quad \ln [L(\pi, \mathbf{n})] = \ln \left(\frac{n!}{\prod_{i=1}^I \prod_{j=1}^J n_{ij}!} \right) + \sum_{i=1}^I \sum_{j=1}^J n_{ij} [\ln(\pi_{i+}) + \ln(\pi_{+j})]$$

$$= \ln \left(\frac{n!}{\prod_{i=1}^I \prod_{j=1}^J n_{ij}!} \right) + \sum_{i=1}^I n_{i+} \ln(\pi_{i+}) + \sum_{j=1}^J n_{+j} \ln(\pi_{+j})$$

and we set the objective function (under H_0)

$$(2.2.10) \quad K(\pi, \lambda_1, \lambda_2) = \ln [L(\pi, \mathbf{n})] + \lambda_1 \left(1 - \sum_{i=1}^I \pi_{i+} \right) + \lambda_2 \left(1 - \sum_{j=1}^J \pi_{+j} \right)$$

This is also called a Lagrange function.

The derivatives of the objective function are given by

$$(2.2.11) \quad \frac{\partial}{\partial \pi_{i+}} K(\pi, \lambda_1, \lambda_2) = \frac{n_{i+}}{\pi_{i+}} - \lambda_1$$

$$\Rightarrow n_{it} = \lambda_1 \pi_{it} \Rightarrow \sum_i n_{it} = n = \lambda_1 \sum_i \pi_{it} \Rightarrow n = \lambda_1$$

$$\frac{\partial}{\partial \pi_{+j}} K(\pi, \lambda_1, \lambda_2) = \frac{n_{+j}}{\pi_{+j}} - \lambda_2$$

$$\frac{\partial}{\partial \lambda_1} K(\pi, \lambda_1, \lambda_2) = 1 - \sum_{i=1}^I \pi_{i+}$$

$$\frac{\partial}{\partial \lambda_2} K(\pi, \lambda_1, \lambda_2) = 1 - \sum_{j=1}^J \pi_{+j}$$

Setting these equal to zero yields

$$(2.2.12) \quad \hat{\lambda}_1 = \hat{\lambda}_2 = n$$

$$\hat{\pi}_{i+} = \frac{n_{i+}}{n}$$

$$\hat{\pi}_{+j} = \frac{n_{+j}}{n}$$

$$\hat{\pi}_{ij} = \hat{\pi}_{i+} \hat{\pi}_{+j} = \frac{n_{i+} n_{+j}}{n^2}$$

$$E_{ij} = n \hat{\pi}_{ij} = \frac{n_{i+} n_{+j}}{n} \quad O_{ij} = n_{ij}$$

We can then compute the X^2 and G^2 statistics. Under the null hypothesis of independence, both statistics are approximately distributed as a Chi-square random variable with degrees-of-freedom given by

$$(2.2.13) \quad df = IJ - 1 - (I - 1) - (J - 1) = (I - 1)(J - 1)$$

where we used the fact that there were $(I - 1)$ free parameters to estimate with $\{\pi_{i+}\}_{i=1}^I$ and $(J - 1)$ free parameters to estimate with $\{\pi_{+j}\}_{j=1}^J$.

Partitioning The Chi-Square Test for Independence

In practice it often turns out that the test for independence is rejected, and in many cases the analysis is ended there. In the next class we will examine log-linear models

which, in general, provide a sophisticated way of examining tables beyond the simple specification of independence. But before discussing log-linear models, we can discuss a way of decomposing the G^2 test of independence into a sum of approximately independent Chi-Square random variables which themselves can give further information about the overall test of independence. This technique is called *Partitioning the Chi-square* test for independence. A partitioning may reveal relationships between certain categories, or groupings of categories, in the table rather than dealing with the table as a whole. It is based upon the reproductive property of the Chi-square distribution which says that if $Z_1^2, \dots, Z_k^2$ are independent Chi-square random variables with one degree-of-freedom then

$$(2.2.14) \quad \sum_{i=1}^k Z_i^2 \sim \chi_k^2$$

We take the following discussion from **Categorical Data Analysis** by Agresti.

“We begin with a single partitioning of G^2 for $2 \times J$ tables, one having $J - 1$ components. The j -th component is identical to G^2 for testing independence in a 2×2 table, where the first column combines 1 through j of the original table, and the second column is column $j + 1$. That is, G^2 for testing independence in a $2 \times J$ table equals a statistic that compares the first two columns, plus a statistic that combines the first two columns and compares them to the third column, ..., plus a statistic that combines the first $J - 1$ columns and compares them to the last column. Each statistic has a single degree-of-freedom.

It might seem more natural to compare G^2 for the $(J - 1)2 \times 2$ tables obtained by pairing each column with a particular one, say the last. These statistics are not asymptotically independent, and their sum does not equal G^2 for testing independence in the full table.

For an $I \times J$ table, asymptotically independent chi-square components result when we compare columns 1 and 2, then combine them and compare them to columns 3,

and so forth. Each of the $J - 1$ statistics has $df = I - 1$. More refined partitioning contain $(I - 1)(J - 1)$ statistics, each having $df = 1$. One such partition, suggested by Lancaster (1949), applies to the $(I - 1)(J - 1)$ 2×2 tables

$$\begin{array}{c|c} \sum_{a < i} \sum_{b < j} n_{ab} & \sum_{a < i} n_{ai} \\ \hline \sum_{b < j} n_{ib} & n_{ij} \end{array}$$

for $i = 2, \dots, I$ and $j = 2, \dots, J$.

Goodman (1968, 1969, 1971b), Irwin (1949), Iverson (1979), and Lancaster (1949) gave rules that help in determining sub-tables for which components of chi-squared are independent. Among these are the following necessary conditions

1. The degrees of freedom for the sub-tables must sum to the degrees of freedom for the original table.
2. Each cell count in the original table must be a cell count in one and only one sub-table.
3. Each marginal total of the original table must be a marginal total for one and only one sub-table.

To check empirically whether a certain partitioning gives independent components, we can note whether the G^2 values for the sub-tables sum to G^2 for the original table."

We next demonstrate an example to illustrate these ideas.

Example 4. The following data is taken from a study by Gallagher (1987) which is also reported in Agresti's book. The table below classifies a sample of psychiatrists by their school of psychiatric thought and by their opinion on the origin of schizophrenia.

Overall Table for Example 4			
School of Psychiatric Thought	Origin of Schizophrenia		
	Biogenic	Environmental	Combination

Eclectic	90	12	78
Medical	13	1	6
Psychoanalytic	19	13	50

The overall test of independence gives

$$G^2 = 23.04 \quad df = (3-1)(3-1) = 4$$

on 4 degrees of freedom which is highly significant since $\chi_4^2(0.01) = 13.28$. We can divide the table into four sub-tables below

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Sub Table 1		
	Biogenic	Environmental
Eclectic	90	12
Medical	13	1
$G^2 = 0.29 \quad df = 1$		

Sub Table 2		
	Biogenic + Environmental	Combination
Eclectic	102	78
Medical	14	6
$G^2 = 1.36 \quad df = 1$		

Sub Table 3		
	Biogenic	Environmental
Eclectic + Medical	103	13
Psychoanalytic	19	13
$G^2 = 12.95 \quad df = 1$		

Sub Table 4		
	Biogenic + Environmental	Combination
Eclectic + Medical	116	84

Medical	32	50
$G^2 = 8.43 \quad df = 1$		

We take the following discussion from Agresti's book

“The first sub-table compares the eclectic and medical schools of psychiatric thought on whether the origin of schizophrenia is biogenic or environmental, given that the classification was in one of these two categories. For this sub-table, $G^2 = 0.29$, based on $df = 1$. The second sub-table compares these two schools in terms of the proportion of times the origin was ascribed to be a combination, rather than biogenic or environmental. This sub-table gives $G^2 = 1.36$, based on $df = 1$. The sum of these first two components equals G^2 for the independence model applied to the first two rows of the (*overall*) table. There is little evidence of a difference between the eclectic and medical schools of thought on the ascribed origin of schizophrenia.

Next we combine the eclectic and medical schools and compare them to the psychoanalytic school. The third sub-table compares them for the (biogenic, environmental) classification, giving $G^2 = 12.95$, with $df = 1$. The fourth sub-table compares them for the (biogenic or environmental, combination) split, giving $G^2 = 8.43$, with $df = 1$.”

The Psychoanalytic school seems more likely than the other schools to ascribe the origins of schizophrenia as being a combination. Of those who chose either the biogenic or environmental origin, members of the psychoanalytic school were somewhat more likely than the other schools to choose the environmental origin. The sum of these four G^2 components equals the value of 23.04 for testing the independence in the original table.

We have discussed the partitioning of the G^2 because an exact partitioning occurs. It turns out that the X^2 statistic for the sub-tables will not sum up to the X^2 for

the overall table, even though the overall X^2 statistic does have an approximate $\chi^2_{(I-1)(J-1)}$ distribution. In practice, it turns out that if a table has small counts, it may be safer to use X^2 instead of G^2 .

Fisher's Exact Test

We have been looking at tests of independence which require estimating a set of unspecified parameters. This happens because we can write the probability of the observed table under the null hypothesis as

$$\begin{aligned}
 (2.2.15) \quad P(\mathbf{n}; \pi) &= \frac{n!}{\prod_{i=1}^I \prod_{j=1}^J n_{ij}!} \prod_{i=1}^I \prod_{j=1}^J (\pi_{i+} \pi_{+j})^{n_{ij}} \\
 &= \frac{n!}{\prod_{i=1}^I \prod_{j=1}^J n_{ij}!} \prod_{i=1}^I \pi_{i+}^{n_{i+}} \prod_{j=1}^J \pi_{+j}^{n_{+j}}
 \end{aligned}$$

We can write the probability mass function for the marginal row totals as

$$(2.2.16) \quad P(n_{1+}, \dots, n_{I+}; \pi) = \frac{n!}{\prod_{i=1}^I n_{i+}!} \prod_{i=1}^I \pi_{i+}^{n_{i+}}$$

and the probability mass function for the column totals as

$$(2.2.17) \quad P(n_{+1}, \dots, n_{+J}; \pi) = \frac{n!}{\prod_{j=1}^J n_{+j}!} \prod_{j=1}^J \pi_{+j}^{n_{+j}}$$

where we used the results of Section 1 about the distribution of collapsed multinomials. It can be shown that if we look at the probability mass function of the table *conditional upon the row and column marginal totals (counts)* can be written

$$(2.2.18) \quad P(\mathbf{n} | n_{1+}, \dots, n_{I+}; n_{+1}, \dots, n_{+J}) = \frac{\prod_{i=1}^I n_i! \prod_{j=1}^J n_{+j}!}{n! \prod_{i=1}^I \prod_{j=1}^J n_{ij}!}$$

which does not depend upon any unknown parameters. Therefore under the null hypothesis the probability distribution is completely determined if we condition on the row and column marginals. For a 2×2 table, this probability can be written

$$(2.2.19) \quad P(n_{11}, n_{12}, n_{21}, n_{22} | n_{1+}, n_{2+}, n_{+1}, n_{+2}) = \frac{\binom{n_{1+}}{n_{11}} \binom{n_{2+}}{n_{21}}}{\binom{n}{n_{+1}}} = \frac{\binom{n_{1+}}{n_{11}} \binom{n_{2+}}{n_{+1} - n_{11}}}{\binom{n}{n_{+1}}}$$

We illustrate this with an example.

Example 5. Consider 16 subjects who complain of chronic headaches. Divide them into two groups of size 8. Let one group receive an aspirin and let the other group receive a placebo. Assume we observe the following table of results

	Aspirin	Placebo	
Pain Reduced	$n_{11} = 7$	$n_{12} = 2$	$n_{1+} = 9$
No Pain Reduction	$n_{21} = 1$	$n_{22} = 6$	$n_{2+} = 7$
	$n_{+1} = 8$	$n_{+2} = 8$	$n = 16$

If column probabilities equal marginal probabilities, then assignment to Aspirin/placebo is same as random within “Pain Reduced”, “No Pain Reduced”.

We will condition on the row total $n_{1+} = 9$ and $n_{2+} = 7$, and test the hypothesis

$$H_0 : \frac{\pi_{11}}{\pi_{+1}} = \frac{\pi_{12}}{\pi_{+2}} \quad \text{versus} \quad H_A : \frac{\pi_{11}}{\pi_{+1}} > \frac{\pi_{12}}{\pi_{+2}}$$

which says that aspirin has the same effect as the placebo, versus the hypothesis that the aspirin is better at relieving pain than the placebo. Under the null hypothesis, the effect of conditioning on the row totals says that all $n_{1+} = 9$ people who experienced pain reduction would have done so no matter which treatment (aspirin or placebo) they were assigned. Therefore when the null hypothesis is true, $n_{1+} = 9$ and $n_{2+} = 7$ are fixed totals, and in this experiment 7 out of the 9 people who would have experienced reduced pain under either treatment just happened to get placed in the aspirin group by the random assignment of people into treatment groups. We can write

$$P(n_{11}, n_{12}, n_{21}, n_{22} | n_{1+}, n_{2+}, n_{+1}, n_{+2}) = \frac{\binom{9}{n_{11}} \binom{7}{8-n_{21}}}{\binom{16}{8}}$$

which is a function of n_{11} . Large values of n_{11} provide evidence against the null hypothesis in favor of the alternative hypothesis. Therefore we can compute the p-value as

$$\begin{aligned} p\text{-value} &= \frac{\binom{9}{7} \binom{7}{8-7}}{\binom{16}{8}} + \frac{\binom{9}{8} \binom{7}{8-8}}{\binom{16}{8}} = \frac{(36)(7)}{12870} + \frac{(9)(1)}{12870} = \frac{261}{12870} = 0.02 \\ &= \Pr(\text{distance observed or more extreme}) \end{aligned}$$

which says that we reject the null hypothesis, and conclude that aspirin does help relieve pain. ■

2.3 Analysis of Ordinal 2-Way Tables

We have been ignoring any possible ordering in the categories of either the row variable or the column variable. This is called treating the variables as *nominal* categorical variables. If there is an ordering to the categories that is important to consider

in the analysis, then we need to consider variables as *ordinal* categorical variables. An example of a situation with ordinal categorical data would be the case where we crossed classified a persons income (classified into income intervals) by rating of job satisfaction (appropriately classified). A test of independence would reveal whether the proportion of people who classify themselves in particular satisfaction categories is the same across all income categories. If the test for independence rejected the null hypothesis, then all we could say is that there is some difference in job satisfaction rating between different income categories. The independence test would not help us answer the question, “Does job satisfaction increase with higher income?”. Note that accepting the null hypothesis of independence does help us answer the question in the negative.

There are many methods for analyzing ordinal categorical data, including several which are available in SAS and other computer programs. For further information on the analysis of ordinal data, see either of Alan Agresti’s books on categorical data analysis.

3. Introduction to Log-Linear Models (Agresti Chapter 6)

In order to introduce general notation for the log-linear modeling approach we will first concentrate on multinomial sampling with 2-Way tables. Remember that we have been writing the probability for an observed table as

$$(3.1) \quad P(\mathbf{n}; \pi) = \frac{n!}{\prod_{i=1}^I \prod_{j=1}^J n_{ij}!} \prod_{i=1}^I \prod_{j=1}^J \pi_{ij}^{n_{ij}}$$

We can also write this as

$$(3.2) \quad P(\mathbf{n}; \pi) = \frac{n!}{\prod_{i=1}^I \prod_{j=1}^J n_{ij}!} \prod_{i=1}^I \prod_{j=1}^J \left(\frac{m_{ij}}{n} \right)^{n_{ij}} = \frac{n! n^{-n}}{\prod_{i=1}^I \prod_{j=1}^J n_{ij}!} \prod_{i=1}^I \prod_{j=1}^J m_{ij}^{n_{ij}}$$

where

$$(3.3) \quad m_{ij} = n\pi_{ij}$$

is the *expected count* in cell i, j . In the form (3.2) we can treat the m_{ij} as parameters instead of the multinomial probabilities π_{ij} . Under the independence hypothesis we can write the expected counts as

$$(3.4) \quad m_{ij} = n\pi_{i+}\pi_{+j} = n \left(\frac{m_{i+}}{n} \right) \left(\frac{m_{+j}}{n} \right) = n^{-1} m_{i+} m_{+j}$$

so the expected cell counts can be written as proportional to the product of the expected marginal counts. By taking logarithms we obtain

$$(3.5) \quad \ln(m_{ij}) = -\ln(n) + \ln(m_{i+}) + \ln(m_{+j})$$

Therefore the log of the expected counts has the appearance of a linear model with main effects for row and column with no interaction. Typically this model is written as

$$(3.6) \quad \ln(m_{ij}) = u + u_{1(i)} + u_{2(j)}$$

or

$$(3.7) \quad \ln(m_{ij}) = \lambda + \lambda_i^X + \lambda_j^Y$$

where the variable X is associated with the rows of the table, and the variable Y is associated with the columns of the table. Under either model, the terms on the right hand side are assumed to follow the usual restrictions

$$(3.8) \quad \sum_{i=1}^I u_{1(i)} = \sum_{j=1}^J u_{2(j)} = 0$$

$$\sum_{i=1}^I \lambda_i^X = \sum_{j=1}^J \lambda_j^Y = 0$$

which implies that there are

$$(3.9) \quad 1 + (I - 1) + (J - 1)$$

free parameters.

We have been looking at the restricted model under the assumption of independence, but we can write the full model, usually called the *saturated model*, as

$$(3.10) \quad \ln(m_{ij}) = u + u_{1(i)} + u_{2(j)} + u_{12(ij)}$$

$$\ln(m_{ij}) = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_{ij}^{XY}$$

where in addition to the restrictions in (3.8), we impose

$$(3.11) \quad \sum_{i=1}^I u_{12(ij)} = \sum_{j=1}^J u_{12(ij)} = 0$$

$$\sum_{i=1}^I \lambda_{ij}^{XY} = \sum_{j=1}^J \lambda_{ij}^{XY} = 0$$

which implies that there are

$$(3.12) \quad (I-1)(J-1)$$

additional free interaction parameters.

The general approach behind log-linear models is to fit *saturated models* of the general form (3.10) and test hypotheses about reduced models, which have natural interpretations, such as the absence of the two way interaction in (3.6) and (3.7) implied independence. A key feature of the kinds of models we will look at is that we will only examine *hierarchical log-linear models*. A log-linear model for a 2-Way table is hierarchical if whenever the terms

$$(3.13) \quad u_{12(ij)} \text{ or } \lambda_{ij}^{XY}$$

are in the model then

$$(3.14) \quad u_{1(i)} \text{ and } u_{2(j)} \quad \text{or} \quad \lambda_i^X \text{ and } \lambda_j^Y$$

are in the model. Note that the term u or λ will always be in the model. Therefore we will not consider models of the form

$$(3.15) \quad \ln(m_{ij}) = u + u_{12(ij)}$$

for example.